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BUSTER

Adaptive Sampling for VLM-guided Unsupervised Video Anomaly Detection

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Motivation

Challenge 1 - Supervision

Frame-wise Human Annotation Is Costly and Difficult to Scale.

Challenge 2 - Computation

Dense VLM Inference Is Costly, While Sparse Sampling Risks Missing Short-Lived Anomalies.

"Can VLMs replace human annotations and guide adaptive sampling for efficient anomaly detection?"

01 THE SAMPLING TRADE-OFF

Sparse sampling may miss anomalies, query-guided sampling adds retrieval overhead, and dense sampling increases computational cost.

INPUT VIDEO

Q: Is there an anomaly in these frames?



LONG STREAM
+10 minutes
sparse vs. dense

SCHEMATIC COMPARISON

Uniform Sampling	SPARSE	Risk of missing brief anomalies.
Query-guided Sampling	RETRIEVE × N	Iterative retrieval and extra latency.
Fixed FPS Sampling	EXPENSIVE	Dense processing across all frames.
BUSTER (ours)	ADAPTIVE	Scout 4 FPS → Burst 8 FPS.

✓ Low cost ✓ Plug-and-play student ✓ Real-time detection

ON-DEMAND VLM OUTPUT — A: "There is Shooting at the Parking lot and People..."

02 HOW BUSTER WORKS

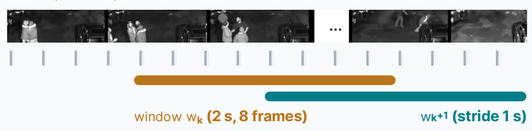
A VLM teacher supplies pseudo-labels; a lightweight student learns anomaly scores **and when to inspect densely**; the VLM reasons only on what the student flags.

A / TEACHER MINING

OFFLINE · LABEL-FREE

A1 Sliding windows @ 4 FPS

$\tau = 2$ s, $\Delta = 1$ s, $n = 8$



Overlapping windows leave **minimal frame omission**: near-continuous temporal coverage at fine resolution, so short anomalies still get a teacher score.

A2 Qwen2.5-VL-32B teacher

"Does any anomaly event occur in this video? Answer Yes or No." **Binary response**

Presence, not category: avoids the VLM's category-label prior and keeps the score grounded in visual context.

A3 Yes/No logit normalization

$$y^{T}_{k} = p_{yes} / (p_{yes} + p_{no} + \varepsilon)$$

confidence without text generation or parsing

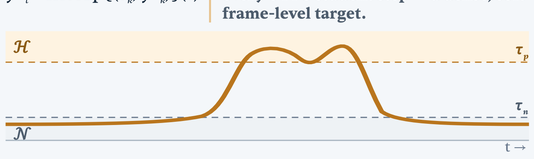
p(Yes) .72 → window score $y^{T}_{k} = 0.72$

p(No) .28

The Yes-share of the binary answer **becomes the window's anomaly score** (0 = normal, 1 = anomalous): one point per window centre c_k.

A4 Pseudo-labels on student timeline

$y^{T}_{t} = \text{Interp}\{(c_k, y^{T}_{k})\}(t)$ — Spreads 1-per-second window scores onto every 8 FPS student step → a dense, soft frame-level target.



$\mathcal{H}$ **High-confidence anomaly set** = $\{t \mid y^{T}_{t} > \tau_p\}$: frames the teacher is **sure are abnormal**.

$\mathcal{N}$ **High-confidence normal set** = $\{t \mid y^{T}_{t} < \tau_n\}$: frames the teacher is **sure are normal**. Ambiguous frames in between are left out of ranking.

Pairs ($a \in \mathcal{H}, n \in \mathcal{N}$) = one sure-anomaly vs one sure-normal frame → feed the ranking loss in Section B (B4) →

→ pseudo-labels y^{T} supervise

B / STUDENT DISTILLATION

SCOUT-AND-BURST CONTROLLER

B1 Frozen LanguageBind

same video @ 8 FPS

$x_t \in \mathbb{R}^{768}$

Compact semantic video features; 8 FPS is a practical trade-off between efficiency and temporal fidelity for 30 FPS video.

B2 Temporal Difference Enc.

$\delta x_t = x_t - x_{t-1}$, $\hat{x}_t = [x_t; \delta x_t]$

semantic change, no optical flow

Captures semantic variation between neighbouring segments: temporal sensitivity without costly motion estimation, a key cue for refining anomaly scores.

B3 Recurrent student · 3.19M

$\hat{y}^{s}_{t} = \sigma(g(h_t))$, $h_t = \text{RNN}(\varphi(x_t), h_{t-1})$

Bi-LSTM (default) · uni-RNN / GRU for causal real-time use

φ → h1 h2 h3 → g·σ

Its score is both the anomaly detector and the control signal for Scout ⇄ Burst.

B4 Rank-aware distillation objective

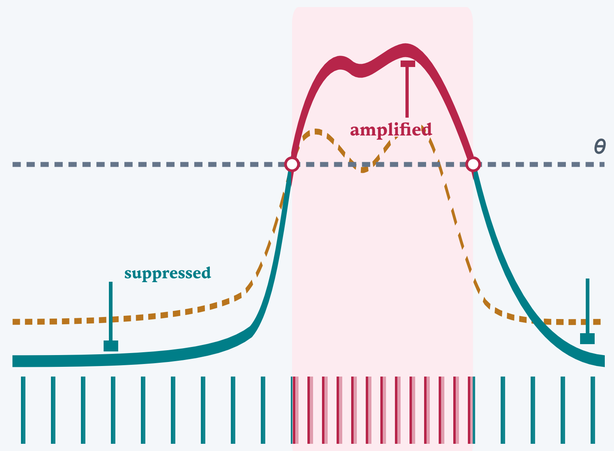
$$\mathcal{L} = \mathcal{L}_{dist} + \lambda_{rank}\mathcal{L}_{rank} + \lambda_{smooth}\mathcal{L}_{smooth}$$

IMITATE	SEPARATE	STABILIZE
$\Sigma (\hat{y}^{s}_{t} - y^{T}_{t})^2$	$\max(0, m - (\hat{y}_a - \hat{y}_n))$	$\mathcal{L}_{smooth}$
match teacher score (MSE)	(a,n) ∈ $\mathcal{H}\times\mathcal{N}$ from A4 · margin m	temporal smoothness of $\hat{y}$

Beyond imitating the teacher, the margin ranking term **enlarges the score gap** between confident anomalous and normal frames, preserving their relative order.

B5 Score-triggered Scout ⇄ Burst

— teacher — student



SCOUT · 4 FPS x_{2u} · sparse watch

$\hat{y} > \theta \rightarrow$ / $\leftarrow \hat{y} \le \theta$

BURST · 8 FPS re-extract interval densely

The student score is the control signal: normal footage stays sparse, and the frame budget moves to suspicious intervals.

53.2% of 8 FPS frames used · same 86.30 AUC as dense 8/8

496.7 FPS detection path · 16.6× real-time at 30 FPS

↓ burst indices ($\hat{y} > \theta$) only → truncated segments ↓

C / BURST REASONING

ON-DEMAND EXPLANATION

C1 Merge into segments

$\mathcal{S} = \{[a_m, e_m]\}_{m=1}^{M}$



Seg 1 · ℓ_1 = 15 s **Seg 2 · ℓ_2 = 8 s**

Consecutive burst indices are merged into candidate segments; M is capped, so the VLM sees a bounded number of clips.

C2 Length-aware budget

$q_m = \min(\ell_m, q_{max}, \max(1, \lfloor B \cdot \ell_m / \Sigma \ell_j \rfloor))$

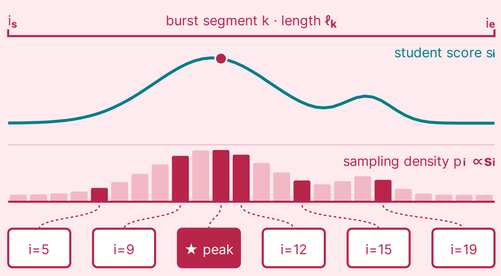
Seg 1 ■■■■■■■■ $q_1 = 8$

Seg 2 ■■■■□□□□ $q_2 = 5$

B = 16 frames total · q_{max} = 8 per segment (illustrative split)

Longer events get more evidence; short ones stay compact — all under one fixed VLM budget.

$p_i = s_i / \Sigma_{j\in[i_s, i_e]} s_j$, $I_k \sim \text{Sample}(q_k; p)$, ★ $t^* = \text{argmax } s$



① **Budget** q_k frames from C2 | ② **Score-aware** high s_i → drawn more often | ③ **Peak kept** closest pick → t^*

Keeps high-confidence cues **and** their surrounding temporal context as a short sequential clip $\tilde{V}_k$ for the VLM.

C4 Qwen2.5-VL-32B on clip $\tilde{V}_m$

objects · environment · interactions · anomalies



MODEL RESPONSE · QWEN2.5-VL-32B

Seg 1 A large **explosion** occurs, engulfing the **area** in **flames and smoke**. **Debris** is **violently propelled into the air** as the **structure collapses**. The **fire intensifies**, consuming the **building and surrounding environment**, leaving it reduced to **rubble**...

Seg 2 **Soldiers crouch and move cautiously** through a **debris-filled area**, indicating **active conflict**. One soldier **collapses to the ground**, apparently injured, while a **civilian runs away from** the **burning building**, showing **panic** and urgency...

RESULTS label-free · Main Paper Tables 1, 2, 4

UCF-Crime AUC (%) · unsupervised

Tur et al.	65.22
GCL	74.20
DyAnNet	79.76
LAVAD (zero-shot)	80.28
BUSTER (offline)	86.30 +6.54
BUSTER (online)	85.50

Across benchmarks (Δ vs best prior unsup.)

	UCF AUC	XD AP	SHT AUC
Offline	86.30 +6.54	80.48 +12.15	78.30 +2.20
Online	85.50 +5.74	80.35 +12.02	75.90

Efficiency · UCF-Crime

	AUC	DATA	FPS
Uniform, 32 frames	63.39	fixed	–
Full 8 FPS	86.30	100%	264.1
BUSTER detection	**86.30**	**53.2%**	**496.7**
BUSTER + VLM (90% normal)	**86.30**	**59.6%**	**351.1**

TEACHER (VLM-ONLY) **82.60** FAR@0.1 15.67%

STUDENT (3.19M) → **86.30** FAR@0.1 0.86%

The distilled student outperforms its VLM teacher.